

KENDRIYA VIDYALAYA SANGATHAN
PATNA REGION
CLASS- X
SESSION – 2024-25
MATHEMATICS STANDARD (041)

SET-B

TIME: 3 hours

MAX.MARKS: 80

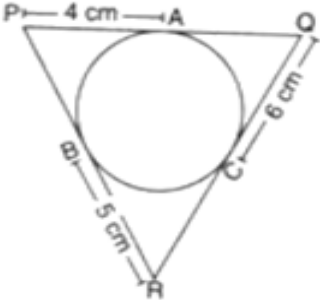
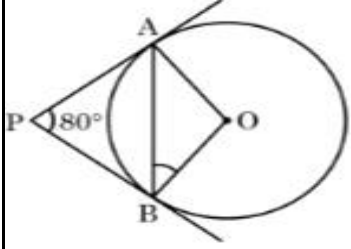
General Instructions:

Read the following instructions carefully and follow them:

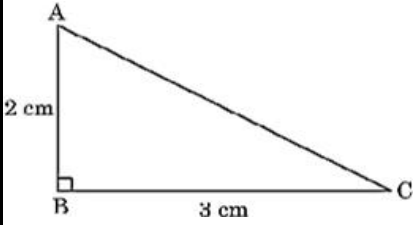
- 1. This question paper contains 38 questions.**
- 2. This Question Paper is divided into 5 Sections A, B, C, D and E.**
- 3. In Section A, Questions no. 1-18 are multiple choice questions (MCQs) and questions no. 19 and 20 are Assertion- Reason based questions of 1 mark each.**
- 4. In Section B, Questions no. 21-25 are very short answer (VSA) type questions, carrying 02 marks each.**
- 5. In Section C, Questions no. 26-31 are short answer (SA) type questions, carrying 03 marks each.**
- 6. In Section D, Questions no. 32-35 are long answer (LA) type questions, carrying 05 marks each.**
- 7. In Section E, Questions no. 36-38 are case study based questions carrying 4 marks each with sub parts of the values of 1, 1 and 2 marks each respectively.**
- 8. All Questions are compulsory. However, an internal choice in 2 Question of Section B, 2 Questions of Section C and 2 Questions of Section D has been provided. An internal choice has been provided in all the 2 marks questions of Section E.**
- 9. Draw neat and clean figures wherever required.**
- 10. Take $\pi = 22/7$ wherever required if not stated.**
- 11. Use of calculators is not allowed.**

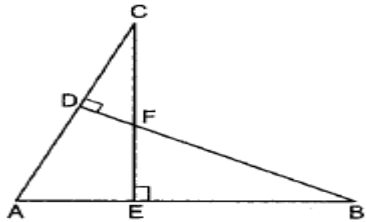
	Section A	
	Section A consists of 20 questions of 1 mark each.	
1.	The zeroes of the polynomial $x^2 - 2x - 3$ are a)-3, 1 b)3, -1 c)3, 1 d)-3, -1	1

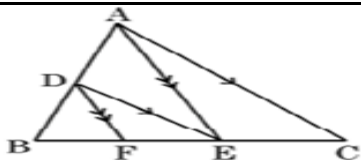
2.	The solution of the pair of equations $x + y = a + b$ and $ax - by = a^2 - b^2$ is: a) $x = -a, y = b$ b) $x = a, y = b$ c) $x = b, y = a$ d) $x = a, y = -b$	1
3.	Which of the following statements is not true? a) A number of secants can be drawn at any point on the circle. b) Only one tangent can be drawn at any point on a circle. c) A chord is a line segment joining two points on the circle d) From a point inside a circle only two tangents can be drawn.	1
4.	If $x + 1, 3x$ and $4x + 2$ are three consecutive terms of an A.P., then the value of x is: a) 5 b) 2 c) 4 d) 3	1
5.	A medicine capsule is in the shape of a cylinder of diameter 0.5 cm with two hemispheres stuck to each of its ends. The length of the entire capsule is 2 cm. The capacity of the capsule is: a) 0.33 cm^3 b) 0.34 cm^3 c) 0.36 cm^3 d) 0.35 cm^3	1
6.	$(\csc\theta - \cot\theta)^2 = ?$ a) $\frac{1+\sin\theta}{1-\sin\theta}$ b) $\frac{1-\cos\theta}{1+\cos\theta}$ c) $\frac{1-\sin\theta}{1+\sin\theta}$ d) $\frac{1+\cos\theta}{1-\cos\theta}$	1

7.	The perimeter of $\triangle PQR$ in the given figure is  a) 15 cm b) 60 cm c) 45 cm d) 30 cm	1
8.	The zeros of the quadratic polynomial $x^2 + 88x + 125$ are a) both negative b) both positive c) both equal d) one positive and one negative	1
9.	The mean and mode of the data are 15 and 12 respectively. The median of the data is a) 11 b) 12 c) 14 d) 15	1
10.	In the given figure, tangents PA and PB from a point P to a circle with centre O are inclined to each other at an angle of 80°. $\angle ABO$ is equal to  a) 50° b) 40° c) 80° d) 100°	1
11.	The roots of the equation $x^2 + 3x - 10 = 0$ are: a) - 2, 5 b) 2, - 5 c) 2, 5	1

	d) - 2, - 5	
12.	If $16 \cot x = 12$, then $\frac{\sin x - \cos x}{\sin x + \cos x}$ equals. a) $\frac{2}{7}$ b) $\frac{3}{7}$ c) 0 d) $\frac{1}{7}$	1
13.	A circus tent is cylindrical to a height of 3 metres and conical above it. If its diameter is 105 m and the slant height of the conical portion is 53 m, calculate the length of the canvas 5 m wide to make the required tent. a) 1800 m b) 2096 m c) 1996 m d) 1947 m	1
14.	One ticket is drawn at random from a bag containing tickets numbered 1 to 40. The probability that the selected ticket has a number which is a multiple of 5 is a) $\frac{1}{3}$ b) $\frac{1}{5}$ c) $\frac{4}{5}$ d) $\frac{3}{5}$	1
15.	If P is a point on x - axis such that its distance from the origin is 3 units, then the coordinates of a point Q on OY such that $OP = OQ$, are a) (0, 0) b) (0, - 3) c) (0, 3) d) (3, 0)	1
16.	Which of the following gives the middle most observation of the data? a) Median b) Mean c) Range d) Mode	1
17.	In the given figure P(5, - 3) and Q(3, y) are the points of trisection of the line segment joining A(7, - 2) and B(1, - 5).  Then, y equals a) $\frac{-5}{2}$ b) 2 c) -4 d) 4	1

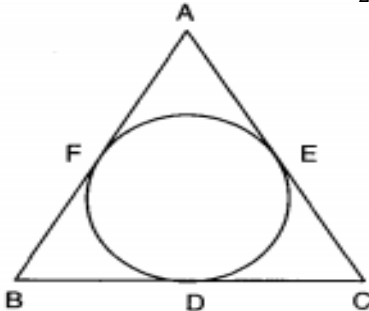
18.	One card is drawn at random from a well shuffled deck of 52 playing cards. The probability that it is a red king is: a) $\frac{2}{26}$ b) $\frac{2}{13}$ c) $\frac{1}{26}$ d) $\frac{1}{52}$	1
	DIRECTION: In the question number 19 and 20, a statement of Assertion (A) is followed by a statement of Reason (R). Choose the correct option A) Both assertion (A) and reason (R) are true and reason (R) is the correct explanation of assertion (A) B) Both assertion (A) and reason (R) are true and reason (R) is not the correct explanation of assertion (A) C) Assertion (A) is true but reason (R) is false. D) Assertion (A) is false but reason (R) is true.	
19.	Assertion (A): The perimeter of $\triangle ABC$ is a rational number. Reason (R): The sum of the squares of two rational numbers is always rational. 	1
20.	Assertion (A): The length of the minute hand of a clock is 7 cm. Then the area swept by the minute hand in 5 minutes is $12\frac{5}{6}cm^2$. Reason (R): The length of an arc of a sector of angle θ and radius r is given by $l = \frac{\theta}{360} \times 2\pi r$	1
	Section B	
	Section B consists of 5 questions of 2 marks each.	

21.	(A) Find H.C.F. and L.C.M. of 56 and 112 by prime factorisation method. OR (B) Check whether 6^n can end with the digit 0 for any natural number n.	2
22.	(A) A box has cards numbered 14 to 99. Cards are mixed thoroughly and a card is drawn from the bag at random. Find the probability that the number on the card, drawn from the box is: a perfect square number a number divisible by 7. OR (B) In a pack of 52 playing cards one card is lost. From the remaining cards , a card is drawn at random. Find the probability the drawn card is queen of heart, if the lost card is a black card.	2
23.	If $x = 30^\circ$,then verify that : $\cos 3x = 4\cos^3 x - 3\cos x$	2
24.	Find the coordinates of a point A, where AB is a diameter of the circle with centre (3, - 1) and the point B is (2, 6).	2
25.	Find the coordinates of the point which divides the join of (- 1, 7) and (4, - 3) in the ratio 2 : 3.	2
	Section C	
	Section C consists of 6 questions of 3 marks each.	
26.	(A) In Fig. if $BD \perp AC$ and $CE \perp AB$, prove that $\triangle AEC \sim \triangle ADB$ $\frac{CA}{AB} = \frac{CE}{DB}$  OR (B) In the given figure, $DE \parallel AC$ and $DF \parallel AE$	3

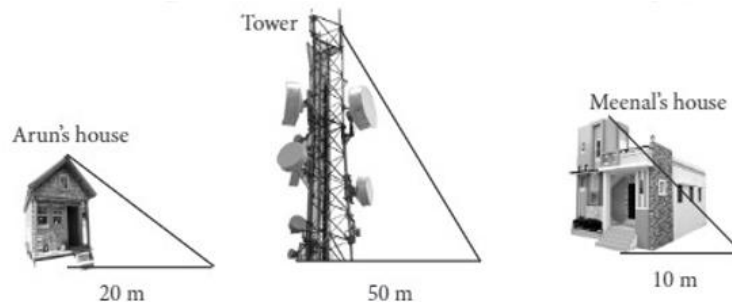


Prove that $\frac{BF}{FE} = \frac{BE}{EC}$

27.	Sum of the areas of two squares is 468 m². If the difference of their perimeters is 24m, find the sides of the squares.	3
28.	If α, β are zeroes of the quadratic polynomial $x^2 + 3x + 2$, find a quadratic polynomial whose zeroes are $(\alpha + 1)$, $(\beta + 1)$.	3
29.	Prove that : $\frac{\sin A}{\sec A + 1} + \frac{\sin A}{\sec A - 1} = 2 \cot A$	3
30.	(A) In a circle of radius 21cm, an arc subtends an angle of 60° at the center. Find: 1. the length of the arc. 2. area of the sector formed by the arc. OR (B) A chord PQ of a circle of radius 10 cm subtends an angle of 60° at the centre of the circle. Find the area of major and minor segments of the circle.	3
31.	Prove that $-\sqrt{5}$ is an irrational number.	3
	Section D	
	Section D consists of 4 questions of 5 marks each	
32.	(A) Check graphically whether the pair of equations $x + 3y = 6$ & $2x - 3y = 12$ is consistent. If so, solve them graphically. OR (B) Points A and B are 70 km. apart on a highway. A car starts from A and another car starts from B simultaneously. If they travel in the same direction, they meet in 7 hours, but if they travel towards each	5

	other, they meet in one hour. Find the speed of the two cars.															
33.	In figure the incircle of $\triangle ABC$ touches the sides BC, CA and AB at D, E and F respectively. Show that $AF + BD + CE = AE + BF + CD = \frac{1}{2}$ (Perimeter of $\triangle ABC$) 	5														
34.	A statue, 1.8 m tall, stands on the top of a pedestal. From a point on the ground, the angle of elevation of top of the statue is 60° and from the same point the angle of elevation of top of the pedestal is 45°. Find the height of the pedestal. (Use $\sqrt{3} = 1.73$)	5														
35.	Find the mean and mode of the following data: <table border="1" data-bbox="373 1382 1181 1688"><thead><tr><th>Weight (in kg)</th><th>Number of students</th></tr></thead><tbody><tr><td>40 - 45</td><td>5</td></tr><tr><td>45 - 50</td><td>11</td></tr><tr><td>50 - 55</td><td>20</td></tr><tr><td>55 - 60</td><td>24</td></tr><tr><td>60 - 65</td><td>28</td></tr><tr><td>65 - 70</td><td>12</td></tr></tbody></table> OR In the following table, Monthly Income (in ₹) of 60 workers given; calculate the median Monthly Income (in ₹).	Weight (in kg)	Number of students	40 - 45	5	45 - 50	11	50 - 55	20	55 - 60	24	60 - 65	28	65 - 70	12	5
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	<table><tr><th>Monthly Income (in ₹)</th><th>No. of Workers</th></tr><tr><td>Income more than ₹ 12000</td><td>60</td></tr><tr><td>Income more than ₹ 15000</td><td>47</td></tr><tr><td>Income more than ₹ 18000</td><td>39</td></tr><tr><td>Income more than ₹ 21000</td><td>21</td></tr><tr><td>Income more than ₹ 24000</td><td>4</td></tr><tr><td>Income more than ₹ 27000</td><td>1</td></tr><tr><td>Income more than ₹ 30000</td><td>0</td></tr></table>	Monthly Income (in ₹)	No. of Workers	Income more than ₹ 12000	60	Income more than ₹ 15000	47	Income more than ₹ 18000	39	Income more than ₹ 21000	21	Income more than ₹ 24000	4	Income more than ₹ 27000	1	Income more than ₹ 30000	0	
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	Section E																	
	Section E consists of 3 case study based questions of 4 marks each.																	
36.	Read the following text carefully and answer the questions that follow: Sehaj Batra gets pocket money from his father everyday. Out of pocket money, he saves money for poor people in his locality. On 1st day he saves ₹ 27.5 On each succeeding day he increases his saving by ₹ 2.5.  1.Find the amount saved by Sehaj on 10th day. 2.Find the amount saved by Sehaj on 25th day. 3.Find the total amount saved by Sehaj in 30 days. OR Find in how many days Sehaj saves ₹ 387.5.	1 1 2																
37.	Meenal was trying to find the height of tower near her house. She is using the properties of similar triangles. The height of Meenal's house is 20 m. When Meenal's house casts a shadow of 10m long on the ground, at the same time, tower casts a shadow of 50 m long and Arun's house casts a shadow of 20 m long on the ground as shown below.																	



Based on the above information, answer the following questions.

1. What is the height of the tower?

- (a) 100m (b) 50m (c) 15m (d) 45m

1

2. What is the height of Arun's house?

- (a) 80m (b) 75m (c) 60m (d) 40m

1

3. If tower casts a shadow of 40m, then find the length of shadow of Arun's house.

2

- (a) 18m (b) 16m (c) 17m (d) 14m

Or

If tower casts a shadow of 40m, then find the length of shadow of Meenal's house.

- (a) 7m (b) 9m (c) 4m (d) 8m

- 38. A wooden toy is shown in the picture. This is a cuboidal wooden block of dimensions $14 \text{ cm} \times 17 \text{ cm} \times 4 \text{ cm}$. On its top there are seven cylindrical hollows for bees to fit in. Each cylindrical hollow is of height 3 cm and radius 2 cm.**



1. Find the volume of wood carved out to make one cylindrical hollow.

1

2. Find the lateral surface area of the cuboid to paint it with green colour.

1

3. Find the volume of wood in the remaining cuboid after carving out seven cylindrical hollows.

2

OR

	Find the surface area of the top surface of the cuboid to be painted yellow.	
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